MIL-A-81083C(OS)
AMENDMENT 3
14 March 1979
SUPERSEDING
AMENDMENT 2
8 January 1971

MILITARY SPECIFICATION

ALTERNATOR-RECTIFIER
(For Torpedo Mk 46 Mods 0, 1, 2, and 4)

This amendment forms a part of Military Specification MIL-A-81083C(OS) dated 2 February 1970.

Wherever it appears in specification delete "Naval Ordnance Systems Command" and substitute "Naval Sea Systems Command".

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*In specification title delete "(For Torpedo Mk 46 Mods 0 and 1)" and substitute "(For Torpedo Mk 46 Mods 0, 1, 2, and 4)".

*Paragraph 2.1, under "DRAWINGS". Add the following:

"DL 1330901	Alternator-Rectifier, and all documents listed thereon (Torpedo Mk 46 Mods 1, 2, and 4)
DL 1330904	Alternator-Rectifier, Shipping (Packing) condition and all documents listed thereon (Torpedo Mk 46 Mods 1, 2, and 4)".

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (501), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1355

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*Paragraph 3.2.2., line 5: Delete "... applicable documents listed in LD 546973, LD 269801, or LM 1330509 ..." and substitute "... applicable documents listed in LD 546973, LD 269801, DL 1330509, or DL 1330901..."

*Paragraph 3.2.3, lines 3 and 4: Delete "...applicable documents listed in LD 546973, LD 269801, or LM 1330509..." and substitute "... applicable documents listed in LD 546973, LD 269801, DL 1330509, or DL 1330901..."

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Paragraph 3.3.1.5: Delete and substitute:

"3.3.1.5 Acceleration. The alternator shall be capable of accelerating from 0 to 24,000 rpm within 0.5 second when loaded as specified in 3.3.1.3.1".

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Paragraph 3.5.1, lines 3 and 4: Delete "The first 2 minutes" and substitute "The first 1 minute"; delete "the remaining 8 minutes" and substitute "the remaining 9 minutes".

*Paragraph 3.6, lines 2 and 3: Delete "...applicable documents listed in LD 546973, LD 269801, or LM 1330509..." and substitute "...applicable documents listed in LD 546973, LD 269801, DL 1330509, or DL 1330901..."

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Paragraph 4.5.3, lines 2 and 3: Delete "Accelerate the alternator to 90 percent of 24,000 rpm within 1.0 plus or minus 0.2 seconds" and substitute "Accelerate the alternator to 24,000 rpm within 0.5 second".

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Table V:

(a) After "Overspeed (4.5.3)" delete "(except that acceleration to 24,000 rpm is to be within 1.0 ± 0.2 seconds)"; add triple asterisk to sample numbers as follows: "7*** 3***"

(b) Add the following note at bottom of table:
"*** Acceleration to 90 percent of 24,000 rpm to be within 1.0 ± 0.2 second".

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*Paragraph 5.1.1, line 3: Delete "...applicable documents listed in LD 269676 or DL 1330516..." and substitute "...applicable documents listed in LD 269676, DL 1330516, or DL 1330904..."

*Paragraph 5.2.1, line 3: Delete "...contained in LD 269676 or DL 1330516..." and substitute "...contained in LD 269676, DL 1330516, or DL 1330904..."

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Replace FIGURE 5 with new FIGURE 5, attached.

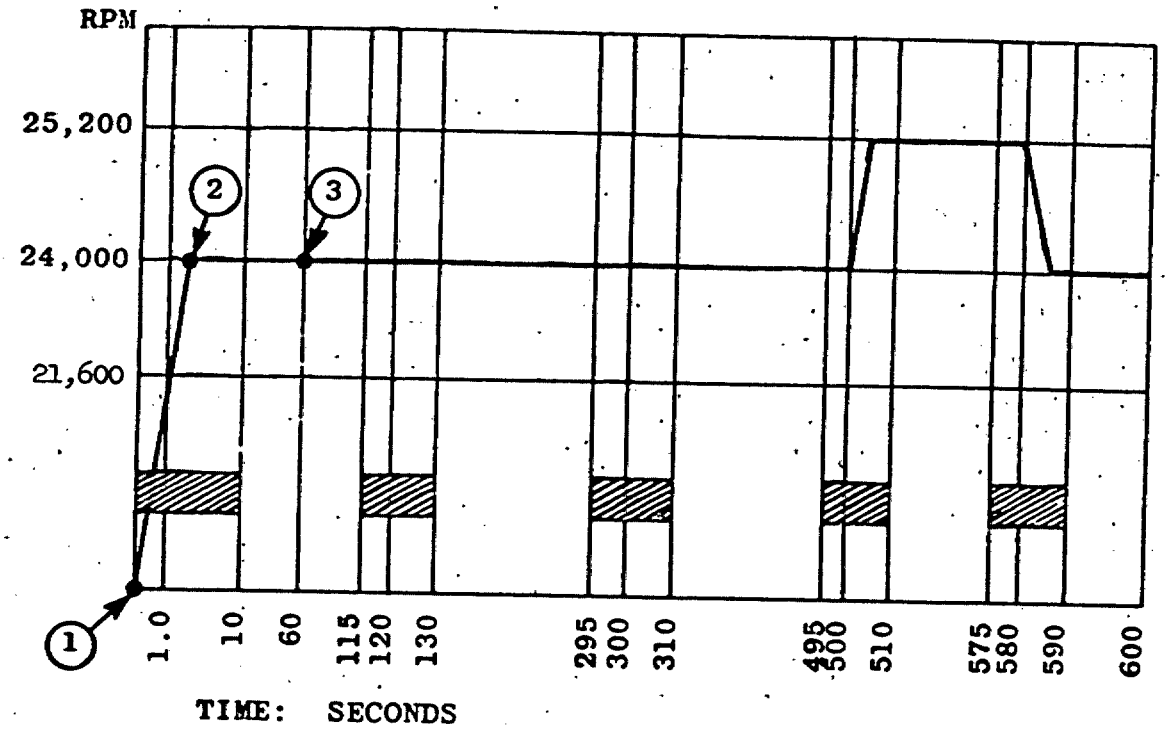
NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Monitor outputs: ±40 vdc steady state, ±40 vdc pulse, and pulse current

Accelerate from zero speed to 90 percent of 24,000 rpm within 1.0 plus or minus 0.2 seconds.

Rate of change of rpm after pulse load is applied shall not exceed 1000 rpm per second.

1. Steady state load applied as defined in paragraph 3.3.1.3.1 and coolant flow-off.

2. (a) Apply pulse loading at rate of one pulse per 0.5 ±0.1 second. Within 5 seconds after reaching 24,000 rpm, load as defined in paragraph 3.3.1.3.2.

   (b) Coolant flow on for operational temperature tests (4.5.4.1).

3. Coolant flow on for operational life tests (4.5.5).

FIGURE 5. Test duty cycle (Service Life - para 3.5.1).

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